Suyoung Lee

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1 Summary

Suyoung Lee is a postdoctoral researcher in the WSP Lab at KAIST whose research lies at the intersection of artificial intelligence (AI) and computer security. His work focuses on identifying emerging attack surfaces across the Web and AI ecosystems and on developing principled, deployable defenses. His research has uncovered over 100 critical flaws in AI-based systems, web browsers, and web applications, resulting in publications in top venues such as IEEE S&P, ACM CCS, USENIX Security, NDSS, WWW, ICML, NeurIPS, and IEEE TDSC. He was selected as one of the Most Valuable Security Researchers by Microsoft Security Response Center (MSRC). He has also served on the program committees of USENIX Security, NDSS, and The ACM Web Conference (WWW), and as a reviewer for ACM Transactions on the Web (TWEB) and ACM Transactions on Software Engineering and Methodology (TOSEM).

2 Employment History

Mar. 2026 – Present **Postdoctoral Researcher**
WSP Lab, KAIST

3 Education

Sept. 2019 – Feb. 2026 **Ph.D., Graduate School of Information Security**
Korea Advanced Institute of Science and Technology (KAIST)
Advisor: Sooel Son

Sept. 2017 – Aug. 2019 **M.S., Graduate School of Information Security**
Korea Advanced Institute of Science and Technology (KAIST)
Advisor: Sooel Son

Mar. 2013 – Aug. 2017 **B.S., Computer Engineering**
Sungkyunkwan University

4 Research Interests

Security, software engineering, and machine learning.

5 Honors and Awards

2025	Cybersecurity Paper Competition by KACS, 2nd place
2022	Best Paper Award (KCC 2022)
2019	Cybersecurity Research Competition by KIISC, 3rd & 4th place
2019	MSRC's 2018–2019 Most Valuable Security Researchers
2015 – 2016	Kwanjeong Educational Foundation Scholarship
2014	Distinguished Freshman Award

6 Publications (*: equal contribution)

6.1 Conference Papers

[C1] Hyunjoon Lee*, **Suyoung Lee***, Heiko Kiesel, and Sooel Son. Overloading Ad Blockers: Exploiting Filtering Logic for Selective Denial of Service. In *Proceedings of the ACM Conference on Computer and Communications Security (CCS)*, 2026 (To appear).

[C2] **Suyoung Lee**, Seongho Keum, Changoo Lee, Dongwon Shin, Sanghyun Hong, Byoungyoung Lee, and Sooel Son. Site Isolation is Dead: How Site Isolation is Broken in Agentic Browsers and Extensions. In *Proceedings of the IEEE Symposium on Security and Privacy (S&P)*, pages 1804–1821, 2026.

[C3] Dongwon Shin, **Suyoung Lee**, Sanghyun Hong, and Sooel Son. You Only Perturb Once: Bypassing (Robust) Ad-Blockers Using Universal Adversarial Perturbations. In *Proceedings of the Annual Computer Security Applications Conference (ACSAC)*, pages 190–206, 2024.

[C4] Byungjoo Kim, **Suyoung Lee**, Seanie Lee, Sooel Son, and Sung Ju Hwang. Margin-based Neural Network Watermarking. In *Proceedings of the International Conference on Machine Learning (ICML)*, pages 16696–16711, 2023.

[C5] Dongwon Shin*, **Suyoung Lee***, and Sooel Son. RICC: Robust Collective Classification of Sybil Accounts. In *Proceedings of the ACM Web Conference (WWW)*, pages 2329–2339, 2023.

[C6] Hoyong Jeong, **Suyoung Lee**, Sung Ju Hwang, and Sooel Son. Learning to Generate Inversion-Resistant Model Explanations. In *Proceedings of the Advances in Neural Information Processing Systems (NeurIPS)*, pages 17717–17729, 2022.

[C7] Gyumin Lim, Gihyuk Ko, **Suyoung Lee**, and Sooel Son. Adversarial Activation based Neural Network Pruning Revision. In *Proceedings of the Korea Computer Congress (KCC)*, 2022.

[C8] **Suyoung Lee**, HyungSeok Han, Sang Kil Cha, and Sooel Son. Montage: A Neural Network Language Model-Guided JavaScript Engine Fuzzer. In *Proceedings of the USENIX Security Symposium (USENIX Security)*, pages 2613–2630, 2020.

[C9] Taekjin Lee, Seongil Wi, **Suyoung Lee**, and Sooel Son. FUSE: Finding File Upload Bugs via Penetration Testing. In *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, 2020.

6.2 Journal Papers

[J1] **Suyoung Lee**, Wonho Song, Suman Jana, Meeyoung Cha, and Sooel Son. Evaluating the Robustness of Trigger Set-Based Watermarks Embedded in Deep Neural Networks. *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 20(4):3434–3448, 2023.

[J2] Gyumin Lim, Gihyuk Ko, **Suyoung Lee**, and Sooel Son. Pruning Deep Neural Networks Neurons for Improved Robustness against Adversarial Examples. *Journal of KIISE*, 50(7):588–597, 2023.

7 Patents

[P1] Sooel Son, Dongwon Shin, and **Suyoung Lee**. Universal Adversarial Attack Method, Apparatus, and Computer Program for Bypassing Machine Learning-based Ad Blocking Systems. Korea Patent 10-2025-0155816, 2025.

[P2] Sooel Son, Dongwon Shin, and **Suyoung Lee**. Random Sampling-based Collective Classification Method and System for Sybil Account Detection. Korea Patent 10-2875522, 2025.

[P3] Sooel Son, Sung Ju Hwang, Hoyong Jeong, and **Suyoung Lee**. Method and Apparatus for Generating Inversion-resistant Model Explanations. Korea Patent 10-2024-0181066, 2024.

[P4] Sooel Son, Dongwon Shin, and **Suyoung Lee**. Random Sampling-based Collective Classification Method and System for Sybil Account Detection. US Patent 18491570, 2023.

[P5] Sooel Son, Gyumin Lim, Gihyuk Ko, and **Suyoung Lee**. Method and Apparatus of Revising a Deep Neural Network for Adversarial Examples. Korea Patent 10-2601761, 2023.

[P6] Sooel Son and **Suyoung Lee**. Evaluating Method of Evaluating Robustness of Artificial Neural Network Watermarking against Model Stealing Attacks. US Patent 17361994, 2021.

[P7] Sooel Son and **Suyoung Lee**. Evaluating Method for the Robustness of Watermarks Embedded in Neural Networks against Model Stealing Attacks. Korea Patent 10-2301295, 2021.

[P8] Sooel Son, Sang Kil Cha, **Suyoung Lee**, Insung Kim, and Taekyu Kim. Method and Apparatus for Testing JavaScript Interpretation Engine using Machine Learning. Korea Patent 10-2132450, 2020.

8 Software Artifacts

2026 A Guardrail System for Detecting and Mitigating Indirect Prompt Injection Attacks
`https://github.com/WSP-LAB/Site-Isolation-Is-Dead`

2024 **YOPO**, A Universal Adversarial Attack Framework against Machine Learning-based Ad-Blockers
`https://github.com/WSP-LAB/YOPO`

2023 **RICC**, A Collective Classification Framework for Finding Sybil Accounts
`https://github.com/WSP-LAB/RICC`

2022 **GNIME**, A Defense Framework against Model Inversion Attacks
`https://github.com/WSP-LAB/GNIME`

2020 **Montage**, A Neural Network Language Model-based JavaScript Engine Fuzzer
`https://github.com/WSP-LAB/Montage`

2020 **FUSE**, A Penetration Testing Tool for Finding File Upload Bugs
`https://github.com/WSP-LAB/FUSE`

9 Professional Services

9.1 Program Committee

2027 USENIX Security, NDSS

2026 WWW (Social Networks and Social Media Track)

9.2 Journal Reviewer

2026 ACM TWEB

2023 ACM TOSEM

10 Invited Talks

Sept. 2026 When New Browser Features Become New Attack Surfaces
UNIST CSE590 Graduate Seminar

July 2026 New Security Threats in Agentic Browsers and Extensions
KIISC AI Security Workshop

11 Teaching

11.1 Teaching Assistant

Spring 2021 IS542 – Web Service Security and Privacy, *KAIST*

Fall 2019 CS492 – Machine Learning Application Trends in Information Security, *KAIST*

Spring 2019 IS542 – Web Service Security and Privacy, *KAIST*

Fall 2018 IS593 – Machine Learning Application Trends in Information Security, *KAIST*